

INTERNET & WEB

The Internet and the World Wide Web (Web) are among the most important developments of the Information Technology Era. They have transformed how people communicate, obtain information, conduct business, learn, work, and interact with digital services. For Electronics Engineering (ECE) and Computer Engineering (CpE) students, understanding the Internet and Web is essential because modern electronic and computing systems increasingly depend on network connectivity.

Although the terms Internet and Web are often used interchangeably, they are not the same. The Internet is the global infrastructure of interconnected computer networks, while the Web is a service that operates on the Internet and allows users to access interconnected documents and applications through web technologies.

1. The Internet: A Global Network

The Internet can be described as a network of networks. It connects millions of private, public, academic, government, and commercial networks around the world.

The Internet allows different devices and networks to communicate because they follow common communication standards and protocols. These standards make it possible for a computer in one country to communicate with a server located thousands of kilometers away.

A simplified communication process is:

User Device → Local Network → Internet Service Provider → Internet → Destination Server

Internet connectivity may be provided through fiber-optic networks, cable, cellular networks, satellite links, Wi-Fi, and other communication technologies.

For engineers, the Internet is important because it demonstrates the integration of electronics, telecommunications, computing, software, and information systems.

2. History and Development of the Internet

The development of the Internet resulted from decades of research in computer communication.

One important predecessor was ARPANET, developed in the United States during the late 1960s. It demonstrated the feasibility of connecting geographically separated computers for communication and resource sharing.

A major development was the use of TCP/IP, a collection of communication protocols that allowed different networks and computers to communicate. TCP/IP became the foundation of the modern Internet.

During the 1990s, the Internet became increasingly accessible to the general public. Commercial Internet service providers, personal computers, web browsers, and websites contributed to rapid growth in Internet use.

The Internet eventually expanded from connecting computers to connecting mobile devices, sensors, vehicles, appliances, industrial machines, and other embedded systems.

3. Internet vs. World Wide Web

The distinction between the Internet and Web is important.

The Internet is the underlying network infrastructure that transports digital information. It supports many services, including email, file transfer, online communication, streaming, cloud computing, and the Web.

The World Wide Web is a system of interconnected digital resources accessed primarily through web browsers.

For example:

- Internet = the communication infrastructure
- Web = one of the services that uses that infrastructure
- Web browser = software used to access web resources
- Website = a collection of related web pages and resources
- Web server = a computer or system that provides web resources

Therefore, the Web depends on the Internet, but the Internet is much broader than the Web.

4. The World Wide Web

The World Wide Web was developed by Tim Berners-Lee at CERN. Its purpose was to make it easier to access and connect information using hypertext.

Three technologies became particularly important to the early Web:

- HTML (HyperText Markup Language) – defines the structure and content of web pages.
- HTTP (HyperText Transfer Protocol) – defines how web resources are requested and transferred.

- URL (Uniform Resource Locator) – identifies the location of a resource on the Web.

A user typically enters a web address or selects a link. The browser sends a request to a web server, which returns the requested resource. The browser then interprets the content and displays it to the user.

5. Web Browsers and Search Engines

A web browser is an application that allows users to access and interact with Web content. Examples include browsers used on computers and mobile devices.

A browser performs several functions, including:

1. Accepting a URL or search request.
2. Resolving the destination's domain name.
3. Establishing communication with the server.
4. Requesting Web resources.
5. Receiving and interpreting HTML, CSS, JavaScript, images, and other content.
6. Displaying the resulting web page or application.

A search engine, on the other hand, helps users discover information available on the Web. It typically uses automated programs to discover and index Web content and algorithms to provide relevant results.

Search engines have become essential information tools, but users must still evaluate the accuracy, credibility, relevance, and date of information they find online.

6. Domain Names and DNS

People generally access websites using domain names such as example.com rather than numerical IP addresses.

The Domain Name System (DNS) translates human-readable domain names into IP addresses that computers can use for network communication.

A simplified process is:

Domain Name → DNS Lookup → IP Address → Web Server → Web Content

DNS is an important part of Internet infrastructure because it provides a practical naming system for the enormous number of connected devices and services on the Internet.

7. Web Technologies

Modern websites are built using several technologies.

HTML provides the structure of a page. CSS (Cascading Style Sheets) controls presentation and visual appearance. JavaScript provides programming and interactive behavior.

Modern Web applications may also communicate with databases, cloud platforms, APIs, authentication systems, and other services.

This has transformed the Web from a collection of static documents into a platform for interactive applications such as:

- Online banking
- E-commerce
- Learning management systems
- Social media
- Online government services
- Video conferencing
- Cloud applications
- Engineering dashboards
- Remote monitoring systems

8. Web Servers, Clients, and Cloud Computing

The traditional Web follows a client-server model. A client, such as a browser, requests information or services from a server.

The server may store web pages, process requests, access databases, authenticate users, or perform computations.

Modern Web systems increasingly use cloud computing, where computing resources, storage, databases, and applications are provided through Internet-connected data centers.

For engineers, cloud services can support applications such as IoT monitoring, remote equipment management, data analytics, artificial intelligence, and large-scale data processing.

9. Internet of Things and the Web

The Internet is no longer limited to traditional computers. The growth of the Internet of Things (IoT) has connected physical devices to networks and Internet services.

An IoT system may contain:

Sensor → Microcontroller/Processor → Network → Cloud/Server → Web Application
→ User

For example, a temperature sensor can collect measurements and transmit them through a network. A server can store and analyze the data, while a Web dashboard allows an engineer to monitor temperature remotely.

This connection between physical devices and Web applications is particularly important to ECE and CpE because it combines electronics, embedded systems, networking, computing, and software.

10. Web 2.0 and Social Interaction

The Web evolved from mostly static information pages into interactive platforms often associated with Web 2.0. Users can create, share, modify, and comment on digital content.

Examples include social networking sites, blogs, collaborative platforms, video-sharing services, and online communities.

This development changed users from being primarily information consumers into content creators and participants.

However, it also introduced challenges such as misinformation, cyberbullying, privacy violations, online scams, excessive screen use, and manipulation through algorithms and targeted content.

11. Internet Security and Privacy

The Internet provides tremendous benefits but also creates security and privacy risks.

Common threats include:

- Phishing
- Malware
- Password attacks
- Identity theft
- Unauthorized access
- Data breaches
- Online fraud
- Malicious websites
- Social engineering

Users should practice basic cybersecurity measures such as using strong and unique passwords, enabling multi-factor authentication when available, keeping software updated, avoiding suspicious links, and protecting personal information.

HTTPS provides encrypted communication between a browser and a website, helping protect information while it is transmitted.

Privacy is another important concern. Users should understand what information they provide online, how services collect and use data, and how their digital footprint can persist over time.

12. The Internet and Philippine Society

In the Philippines, Internet connectivity has become increasingly important in education, business, government, healthcare, banking, entertainment, and communication.

Online platforms support distance learning, digital payments, e-commerce, government services, remote work, telemedicine, disaster communication, and social interaction.

At the same time, unequal access to reliable and affordable Internet creates a digital divide. Differences in connectivity, devices, digital skills, and access to information can affect educational and economic opportunities.

Engineers therefore have an important role in developing communication infrastructure and technologies that are reliable, accessible, secure, energy-efficient, and appropriate for local conditions.

13. The Future of the Internet and Web

The Internet continues to evolve toward increasingly intelligent and connected environments. Technologies such as 5G and future communication systems, edge computing, artificial intelligence, IoT, cloud computing, augmented reality, virtual reality, and Web-based AI applications are changing how people and machines interact.

The future Web is likely to become more personalized, intelligent, immersive, and integrated with physical environments.

For ECE and CpE students, this means that future engineering systems may no longer be designed as isolated devices. Instead, they will increasingly be part of interconnected ecosystems involving sensors, embedded processors, communication networks, cloud platforms, AI systems, and human users.